THE BRANE COHOMOLOGY CONJECTURE FOR EXCEPTIONAL SUPER YANG–MILLS IS FALSE

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Abstract. Rios, Marrani and Chester asserted that the spinors of exceptional super Yang–Mills theory in $(8n+3)+3$ dimensions are the cohomology of its largest brane. We prove this false for every $n \geq 1$: on spinors the Casimir operator of the brane rotation algebra $\mathfrak{so}_m$, $m = 4n+3$, equals $m(m-1)/8$, which is never an integer, while on forms it equals $k(m-k)$.

1. Introduction

Rios, Marrani and Chester proposed in [11] a family of supersymmetric Yang–Mills theories in the signatures $(9+8n)+1$, $(10+8n)+2$, $(11+8n)+3$ and $(12+8n)+4$, which they called exceptional, by moving $4n$ timelike directions of Nishino's signatures [10] into space; supersymmetry in these dimensions goes back to [10, 13]. In [12] they took up the case $27+3$, in which the supersymmetry algebra

$$\{Q_\alpha, Q_\beta\} = (\gamma^{\mu\nu\rho})_{\alpha\beta} Z_{\mu\nu\rho} + (\gamma^{\mu_1\cdots\mu_7})_{\alpha\beta} Z_{\mu_1\cdots\mu_7} + (\gamma^{\mu_1\cdots\mu_{11}})_{\alpha\beta} Z_{\mu_1\cdots\mu_{11}} + (\gamma^{\mu_1\cdots\mu_{15}})_{\alpha\beta} Z_{\mu_1\cdots\mu_{15}} \tag{1}$$

carries the charges of a 3-, 7-, 11- and 15-brane. That these theories cannot be built by repeating Nishino's construction is shown in [2]; related geometric questions for Yang–Mills fields and for compactifications are treated in [4, 3]. The present paper is about a different claim of [12], made in its Section III and intended to extend the observation of Dijkgraaf, Verlinde and Verlinde [5] that the sixteen central charges of the five-brane are fluxes through its homology cycles.

The claim reads as follows. Write $\Lambda^k\mathbf{15}$ for the k-th exterior power of the vector representation of $\mathfrak{so}_{15}$. Equations (3.4) and (3.5) of [12] state that

$$\underbrace{\mathbf{16384}}_{\mathfrak{so}_{27,3}\text{ MW spinor}} = \bigoplus_i \Lambda^{2i+1}\mathbf{15}, \qquad \underbrace{\mathbf{16384'}}_{\mathfrak{so}_{27,3}\text{ MW conj. spinor}} = \bigoplus_i \Lambda^{2i}\mathbf{15}, \tag{2}$$

covariantly under $\mathfrak{so}_{15}$, and call the right-hand sides the odd and even cohomology of the 15-brane, the largest brane in (1). The text before (3.4) says that the semispinor **32768** of $\mathfrak{so}_{28,4}$ is the total cohomology of the same brane, and (3.6)–(3.8) assert, as "a general fact", that for every level n the semispinor $\mathbf{2}^{4n+3}$ of $\mathfrak{so}_{8n+4,4}$ is the total cohomology of a $(4n+3)$-brane; for $n=1$ the **128** of $\mathfrak{e}_{8(-24)}$ is said to be the total cohomology of the 7-brane.[1]

Dimension counts cannot decide such a statement, since many modules share a dimension. The question is whether some subalgebra of $\mathfrak{so}_{s,t}$ isomorphic to $\mathfrak{so}_{4n+3}$ makes the spinor module isomorphic to a sum of exterior powers of its vector module. For the five-brane of [5] it does: there $\mathfrak{so}_5 \subset \mathfrak{gl}_5 \subset \mathfrak{so}_{5,5}$ acts on $\mathbb{R}^{5,5} = \mathbb{R}^5 \oplus (\mathbb{R}^5)^*$, and the spinors of $\mathfrak{so}_{5,5}$ restrict to $\Lambda^\bullet\mathbb{R}^5$ up to a power of the determinant [6, §20.1]. We show that in the signatures of [11, 12] it never does when $\mathfrak{so}_{4n+3}$ is the compact rotation algebra of a spatial worldvolume, and in the signatures

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[1] The dimensions agree: $\sum_k \binom{15}{k} = 2^{15}$ and $\sum_{k\ \text{odd}} \binom{15}{k} = 2^{14}$. The list printed in (3.5) of [12] ends in $\mathbf{25'}$, which should be $\binom{15}{14} = 15$; with 25 the list adds up to 16394.

$(8n+3)+3$ with $n \geq 2$ it never does for any real form. Since [12] presents the identification as a general fact without proof, we refer to it as the brane cohomology conjecture.

Theorem A. *Let $n \geq 2$ and $m = 4n+3$.*

(a) *Let $\mathfrak{h} \subset \mathfrak{so}_{8n+3,3}$ be a subalgebra isomorphic to a real form of $\mathfrak{so}_m(\mathbb{C})$. No exterior power of the vector module of $\mathfrak{h}$ occurs in either semispinor of $\mathfrak{so}_{8n+3,3}$, even after complexification.*

(b) *Let $\mathfrak{h} \subset \mathfrak{so}_{8n+4,4}$ be a subalgebra isomorphic to the compact algebra $\mathfrak{so}_m$. No exterior power of the vector module of $\mathfrak{h}$ occurs in either semispinor of $\mathfrak{so}_{8n+4,4}$, even after complexification.*

For $n = 3$ this says that (2) is false for every choice of $\mathfrak{so}_{15}$ inside $\mathfrak{so}_{27,3}$, and that the **32768** of $\mathfrak{so}_{28,4}$ is not the cohomology of a 15-brane; for $n = 2$ it says the same about $19+3$ and $20+4$. The case $n = 1$ needs a separate argument, because $\mathfrak{so}_7$ has an eight-dimensional spin module that fits into $\mathbb{R}^{12,4}$.

Theorem B. *Let $\mathfrak{h}$ be a subalgebra of $\mathfrak{so}_{11,3}$ or of $\mathfrak{so}_{12,4}$ isomorphic to the compact algebra $\mathfrak{so}_7$. Neither semispinor contains $\Lambda^2\mathbb{R}^7$, $\Lambda^3\mathbb{R}^7$, $\Lambda^4\mathbb{R}^7$ or $\Lambda^5\mathbb{R}^7$, even after complexification. In particular the* **128** *of $\mathfrak{so}_{12,4}$ is not $\Lambda^\bullet\mathbb{R}^7$ and the* **64** *of $\mathfrak{so}_{11,3}$ is not $\bigoplus_i \Lambda^{2i+1}\mathbb{R}^7$.*

The obstruction is the signature. Over $\mathbb{C}$ the identification holds, through $\mathfrak{so}_m(\mathbb{C}) \subset \mathfrak{gl}_m(\mathbb{C}) \subset \mathfrak{so}_{2m}(\mathbb{C})$, and Remark 5.2 shows that it also holds for the noncompact real form $\mathfrak{so}_{4n+1,2}$ inside $\mathfrak{so}_{8n+4,4}$; but the rotations of the spatial worldvolume of a brane form a compact algebra, and for $\mathfrak{so}_{8n+3,3}$ no real form helps. The proofs rest on one computation, Lemma 2.1: on every spinor the Casimir operator of $\mathfrak{so}_m$ equals $m(m-1)/8$, which is not an integer when $m \equiv 3 \pmod 4$, while on Λ^k it equals the integer $k(m-k)$ (Figure 1).

Section II of [12] also introduces a "dual M-theory" in $13+1$ dimensions as the worldvolume theory of the magnetic dual of the 10-brane, obtained by reduction from $26+2$ to $26+1$.

Proposition C. *In the conventions of* [12]*, the reduction $26+2 \to 26+1$ takes the 14-brane with worldvolume $14+2$ to a 14-brane with worldvolume $14+1$, not to the 13-brane with worldvolume $13+1$; and the 10-brane and the 14-brane are not electric–magnetic duals in $26+1$ dimensions.*

Section 2 fixes notation, Section 3 bounds the small modules of $\mathfrak{so}_{2l+1}$, Section 4 shows how such an algebra can sit in $\mathfrak{so}_{s,t}$, Section 5 proves Theorems A and B, and Section 6 proves Proposition C. The finite arithmetic used below is rechecked by a short program in the archive of this paper [1]; no proof depends on it.

2. Preliminaries

2.1. **Clifford algebras and semispinors.** Let $V = \mathbb{R}^{s,t}$ with the metric $\eta(x,x) = x_1^2 + \cdots + x_s^2 - x_{s+1}^2 - \cdots - x_{s+t}^2$, and let $V_\mathbb{C} = V \otimes \mathbb{C}$ with the complex bilinear extension of η. The Clifford algebra $\mathrm{Cl}(V_\mathbb{C})$ is generated by $V_\mathbb{C}$ subject to

$$vw + wv = 2\eta(v,w) \qquad (v, w \in V_\mathbb{C}). \tag{3}$$

Suppose $s+t = 2r$ is even. Over $\mathbb{C}$ there is a basis $e_1, \dots, e_{2r}$ of $V_\mathbb{C}$ with $\eta(e_a, e_b) = \delta_{ab}$ (multiply the timelike vectors of a real orthonormal basis by i). Then $\mathrm{Cl}(V_\mathbb{C}) \cong \mathrm{End}(S)$ for a unique irreducible module S of dimension 2^r [9, Ch. I, §§4–5]. Put $\Gamma = i^r e_1 e_2 \cdots e_{2r}$. Moving each factor past the others with (3) gives $(e_1 \cdots e_N)^2 = (-1)^{N(N-1)/2}$ for $N = 2r$, that is $(-1)^{r(2r-1)} = (-1)^r$, and $(i^r)^2 = (-1)^r$, so $\Gamma^2 = 1$. Since N is even, Γ anticommutes with every e_a and commutes with every product $e_a e_b$. The semispinors are the eigenspaces $S^\pm = \{\psi \in S : \Gamma\psi = \pm\psi\}$; the invertible element e_1 exchanges them, so $\dim S^\pm = 2^{r-1}$.

For $v, w \in V_{\mathbb{C}}$ write $E_{v,w} \in \mathfrak{so}(V_{\mathbb{C}})$ for the map $E_{v,w}(x) = \eta(w,x)\,v - \eta(v,x)\,w$. The elements E_{e_a,e_b} with $a < b$ form a basis of $\mathfrak{so}(V_{\mathbb{C}})$. If $\eta(v,w) = 0$, then $vw = -wv$ by (3), and for every $x \in V_{\mathbb{C}}$ two applications of (3) give

$$vwx = v\big(-xw + 2\eta(w,x)\big) = -(vx)w + 2\eta(w,x)v = xvw - 2\eta(v,x)w + 2\eta(w,x)v,$$

so that

$$\big[\tfrac{1}{2}vw,\, x\big] = \tfrac{1}{2}(vwx - xvw) = \eta(w,x)\,v - \eta(v,x)\,w = E_{v,w}(x). \tag{4}$$

Hence the linear map $\sigma : \mathfrak{so}(V_{\mathbb{C}}) \to \mathrm{Cl}(V_{\mathbb{C}})$ with $\sigma(E_{e_a,e_b}) = \frac{1}{2}e_a e_b$ satisfies $[\sigma(X), x] = Xx$ for $X \in \mathfrak{so}(V_{\mathbb{C}})$ and $x \in V_{\mathbb{C}}$; it is a homomorphism of Lie algebras and gives the spin representation of $\mathfrak{so}(V_{\mathbb{C}})$ on S [9, Ch. I, §6]. Each $\sigma(X)$ is a sum of products of two vectors, so it commutes with Γ and preserves S^+ and S^-.

A Majorana–Weyl spinor of $\mathfrak{so}_{s,t}$ is a real form of S^+ or S^-; it exists when $s - t \equiv 0 \pmod 8$ [9, Ch. I, §§4–5], as in all signatures considered here. If $S^{\pm}_{\mathbb{R}}$ is such a real form and $\mathfrak{h} \subset \mathfrak{so}_{s,t}$ is a subalgebra, then a nonzero $\mathfrak{h}$-module map $f : M \to S^{\pm}_{\mathbb{R}}$ from a real $\mathfrak{h}$-module M gives the nonzero $\mathfrak{h} \otimes \mathbb{C}$-module map $f \otimes \mathbb{C} : M \otimes \mathbb{C} \to S^{\pm}$. It therefore suffices to prove Theorems A and B for the complex semispinors $S^{\pm}$ and the complexified modules $\Lambda^k(M \otimes \mathbb{C})$, which is the meaning of "even after complexification". Since all modules below are completely reducible, a module occurs as a submodule exactly when it occurs as a quotient.

2.2. **The Casimir element.** Let $U \subset V_{\mathbb{C}}$ be a subspace of dimension m on which η is nondegenerate, with a basis $u_1, \dots, u_m$ satisfying $\eta(u_i, u_j) = \delta_{ij}$. Then $V_{\mathbb{C}} = U \oplus U^{\perp}$. Write $E_{ij} = E_{u_i,u_j}$. The E_{ij} with $i < j$ form a basis of $\mathfrak{so}(U)$, which we view inside $\mathfrak{so}(V_{\mathbb{C}})$ as the maps that are zero on $U^{\perp}$. From the definition,

$$E_{ij}u_j = u_i, \qquad E_{ij}u_i = -u_j, \qquad E_{ij}u_l = 0 \quad (l \notin \{i,j\}), \qquad E_{ij} = 0 \text{ on } U^{\perp}. \tag{5}$$

Define

$$\Omega = -\sum_{1 \le i < j \le m} E_{ij}^2 \tag{6}$$

in the universal enveloping algebra of $\mathfrak{so}(U)$.

Lemma 2.1. (o) *Ω is central in the enveloping algebra of $\mathfrak{so}(U)$. Hence it acts by a scalar on every irreducible $\mathfrak{so}(U)$-module, and every $\mathfrak{so}(U)$-module map intertwines the actions of Ω.*

(i) *On $\Lambda^k U$ the element Ω acts by $k(m-k)$.*

(ii) *On S, and hence on S^+ and S^-, the element Ω acts by $m(m-1)/8$.*

(iii) *If $m \equiv 3 \pmod 4$, then $m(m-1)/8 \neq k(m-k)$ for every integer k.*

Proof. (o) For $X, Y \in \mathfrak{so}(U)$ put $B(X,Y) = -\frac{1}{2}\operatorname{tr}_U(XY)$. Because the trace is cyclic,

$$B([Z,X],Y) = -B(X,[Z,Y]) \qquad (X,Y,Z \in \mathfrak{so}(U)).$$

By (5), $E_{ij}E_{kl}$ has a nonzero diagonal entry only if $\{i,j\} = \{k,l\}$, and E_{ij}^2 is -1 on u_i and u_j and 0 on the other u_l; hence $B(E_{ij}, E_{kl}) = \delta_{ik}\delta_{jl}$ for $i < j$ and $k < l$. Number the E_{ij} with $i < j$ as $X_1, \dots, X_N$, so $\Omega = -\sum_a X_a^2$. For $Y \in \mathfrak{so}(U)$ write $[Y, X_a] = \sum_b c_{ab}X_b$ with $c_{ab} = B([Y,X_a],X_b)$. Invariance gives $c_{ab} = -B(X_a,[Y,X_b]) = -c_{ba}$, and so

$$\Big[Y, \sum_a X_a^2\Big] = \sum_a \big([Y,X_a]X_a + X_a[Y,X_a]\big) = \sum_{a,b} c_{ab}(X_bX_a + X_aX_b) = 0,$$

because the summand is antisymmetric in (a,b) while $X_bX_a + X_aX_b$ is symmetric. Thus Ω commutes with $\mathfrak{so}(U)$, hence with its enveloping algebra. On an irreducible module it acts by a

scalar by Schur's lemma, and a module map f satisfies $f \circ \Omega = \Omega \circ f$ because Ω is a polynomial in the action.

(i) Each E_{ij} acts on $\Lambda^k U$ as a derivation: $E_{ij}(x_1 \wedge \cdots \wedge x_k) = \sum_r x_1 \wedge \cdots \wedge E_{ij}x_r \wedge \cdots \wedge x_k$. Let $u_I = u_{i_1} \wedge \cdots \wedge u_{i_k}$ with $I = \{i_1 < \cdots < i_k\}$; these form a basis of $\Lambda^k U$. Fix $i < j$.

- If $i, j \notin I$, every term vanishes by (5), so $E_{ij}u_I = 0$.
- If $i, j \in I$, the derivation produces two terms: one in which u_j is replaced by u_i, which then occurs twice, and one in which u_i is replaced by $-u_j$, which then occurs twice. Both vanish, so $E_{ij}u_I = 0$.
- If $i \in I$ and $j \notin I$, then $E_{ij}u_I = -u_{I'}$, where $u_{I'}$ is u_I with the factor u_i replaced by u_j in the same position. Now $u_{I'}$ contains u_j and not u_i, so $E_{ij}u_{I'}$ is $u_{I'}$ with that factor replaced by u_i, which is u_I. Hence $E_{ij}^2 u_I = -u_I$. The case $j \in I$, $i \notin I$ is the same with the signs exchanged, and again $E_{ij}^2 u_I = -u_I$.

Therefore $\Omega u_I = \#\{\{i,j\} : |\{i,j\} \cap I| = 1\} \cdot u_I$. A pair meeting I in exactly one element is a choice of one element of I and one of its complement, so there are $k(m-k)$ of them, and $\Omega = k(m-k)$ on $\Lambda^k U$.

(ii) By (4), E_{ij} acts on S as multiplication by $\frac{1}{2}u_i u_j$. For $i \neq j$, (3) gives $u_j u_i = -u_i u_j$ and $u_i^2 = u_j^2 = 1$, so

$$\left(\tfrac{1}{2}u_i u_j\right)^2 = \tfrac{1}{4}\, u_i(u_j u_i)u_j = -\tfrac{1}{4}\, u_i^2 u_j^2 = -\tfrac{1}{4}.$$

Hence each of the $\binom{m}{2}$ terms of (6) contributes $+\frac{1}{4}$, and $\Omega = \frac{1}{4}\binom{m}{2} = \frac{m(m-1)}{8}$ on all of S, so on S^+ and on S^-.

(iii) If $m = 4n+3$, then $m(m-1)/8 = (4n+3)(4n+2)/8 = (4n+3)(2n+1)/4$. The numerator is a product of two odd numbers, so the quotient is not an integer, while $k(m-k)$ is an integer. □

For the three values of m that occur most often below, the two sets of values are

m	$k(m-k),\ 0 \le k \le m$	$m(m-1)/8$
7	0, 6, 10, 12	21/4
11	0, 10, 18, 24, 28, 30	55/4
15	0, 14, 26, 36, 44, 50, 54, 56	105/4

(each value of $k(m-k)$ is attained at k and $m-k$); see also Figure 1.

3. Small modules of $\mathfrak{so}_{2l+1}$

Let $l \ge 1$ and $W = \mathbb{C}^{2l+1}$ with an orthonormal basis, and let $\mathfrak{t}$ be the Cartan subalgebra of $\mathfrak{so}(W) \cong \mathfrak{so}_{2l+1}(\mathbb{C})$ spanned by l commuting rotations in disjoint coordinate planes. With $e_1, \ldots, e_l$ the corresponding basis of $\mathfrak{t}^*$ and the standard inner product $(e_i, e_j) = \delta_{ij}$, the weights of W are $\pm e_1, \ldots, \pm e_l, 0$, the positive roots are

$$e_i - e_j, \quad e_i + e_j \quad (1 \le i < j \le l), \qquad e_i \quad (1 \le i \le l),$$

the half sum of the positive roots is $\rho = \sum_{i=1}^{l}\left(l - i + \frac{1}{2}\right)e_i$, and the fundamental weights are $\omega_i = e_1 + \cdots + e_i$ for $i < l$ and $\omega_l = \frac{1}{2}(e_1 + \cdots + e_l)$. The irreducible module $V(\omega_i)$ is $\Lambda^i W$ for $1 \le i < l$, the module $\Lambda^l W$ is irreducible with highest weight $e_1 + \cdots + e_l = 2\omega_l$ [6, §19.4], and $V(\omega_l)$ is the spin module, of dimension 2^l [6, §20.1]. The Weyl dimension formula [6, §24.1] reads

$$\dim V(\lambda) = \prod_{\alpha > 0} \frac{(\lambda + \rho, \alpha)}{(\rho, \alpha)}. \tag{7}$$

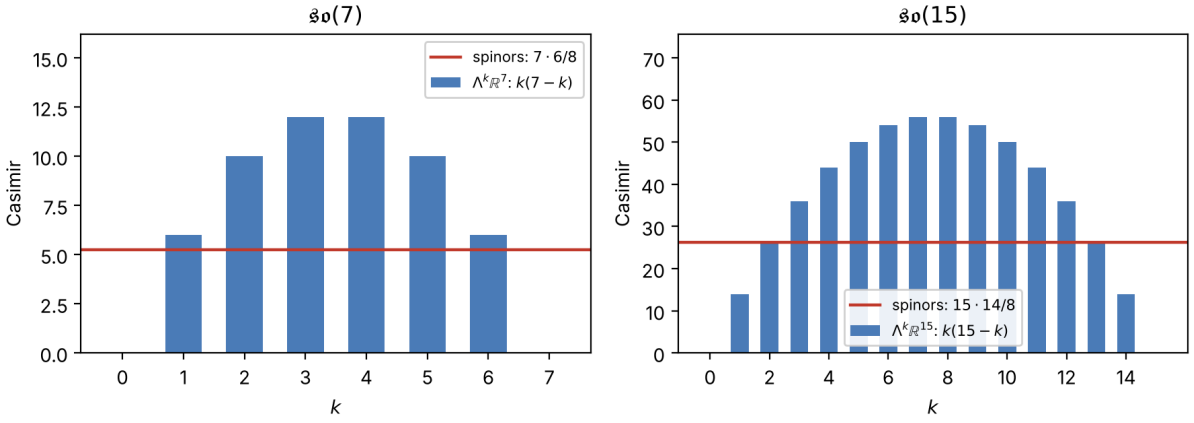


FIGURE 1. The Casimir operator (6) on the exterior powers $\Lambda^k\mathbb{R}^m$ (bars) and on spinors (line), for $m = 7$ and $m = 15$. The line meets no bar.

If λ and μ are dominant, then $(\mu, \alpha) \geq 0$ for every positive root, so each factor of (7) for $\lambda + \mu$ is at least the corresponding factor for λ, and all factors are positive. Hence

$$\dim V(\lambda + \mu) \geq \dim V(\lambda) \qquad (\lambda, \mu \text{ dominant}). \tag{8}$$

We need two dimensions. In (7) only the roots with a nonzero e_1-coordinate see a change of the first coordinate.

- $\lambda = 2\omega_1 = 2e_1$. Here $(\rho, e_1) = l - \frac{1}{2}$, $(\rho, e_1 - e_j) = j - 1$ and $(\rho, e_1 + e_j) = 2l - j$, and adding $2e_1$ raises each by 2. So

$$\dim V(2\omega_1) = \frac{2l+3}{2l-1} \prod_{j=2}^{l} \frac{j+1}{j-1} \prod_{j=2}^{l} \frac{2l-j+2}{2l-j} = \frac{2l+3}{2l-1} \cdot \frac{l(l+1)}{2} \cdot \frac{2l(2l-1)}{l(l+1)} = l(2l+3),$$

  the two products telescoping.
- $\lambda = \omega_1 + \omega_l$, compared with ω_l. The vector $\omega_l + \rho$ has coordinates $l, l-1, \ldots, 1$, and $\lambda + \rho$ has first coordinate $l + 1$ and the others unchanged. So

$$\frac{\dim V(\omega_1 + \omega_l)}{\dim V(\omega_l)} = \frac{l+1}{l} \prod_{j=2}^{l} \frac{j}{j-1} \prod_{j=2}^{l} \frac{2l-j+2}{2l-j+1} = \frac{l+1}{l} \cdot l \cdot \frac{2l}{l+1} = 2l,$$

  and $\dim V(\omega_1 + \omega_l) = 2l \cdot 2^l$.

Lemma 3.1. *Let $l \geq 3$. Every irreducible $\mathfrak{so}_{2l+1}(\mathbb{C})$-module of dimension less than $l(2l+1)$ is trivial, isomorphic to W, or isomorphic to the spin module, of dimension 2^l.*

Proof. Let $\lambda = \sum_i a_i\omega_i$ be the highest weight, with integers $a_i \geq 0$.

If $a_i \geq 1$ for some i with $2 \leq i \leq l-1$, then by (8) with $\mu = \lambda - \omega_i$,

$$\dim V(\lambda) \geq \dim V(\omega_i) = \binom{2l+1}{i} \geq \binom{2l+1}{2} = l(2l+1),$$

because $\binom{2l+1}{i}$ increases with i for $i \leq l$.

Otherwise $\lambda = a\omega_1 + b\omega_l$. If $a \geq 2$, then $\dim V(\lambda) \geq \dim V(2\omega_1) = l(2l+3) > l(2l+1)$. If $b \geq 2$, then $\dim V(\lambda) \geq \dim V(2\omega_l) = \dim \Lambda^l W = \binom{2l+1}{l} \geq \binom{2l+1}{2} = l(2l+1)$, using $l \geq 2$. If $a = b = 1$, then $\dim V(\lambda) = 2l \cdot 2^l = l \cdot 2^{l+1} \geq l(2l+1)$, since $2^{l+1} \geq 2l+1$. The remaining weights are 0, ω_1 and ω_l, which give the trivial module, W and the spin module. □

Corollary 3.2. *Let $m = 4n+3$ and $W = \mathbb{C}^m$. If $n \geq 2$, every $\mathfrak{so}_m(\mathbb{C})$-module of dimension at most $8n+8$ is a direct sum of trivial modules and copies of W. If $n = 1$, every $\mathfrak{so}_7(\mathbb{C})$-module of dimension at most 16 is a direct sum of trivial modules, copies of $W = \mathbb{C}^7$ and copies of the spin module $\mathbb{C}^8$.*

Proof. Finite-dimensional modules of the semisimple algebra $\mathfrak{so}_m(\mathbb{C})$ are completely reducible, so it suffices to treat irreducible modules. Here $m = 2l+1$ with $l = 2n+1 \geq 3$, and

$$l(2l+1) - (8n+8) = (2n+1)(4n+3) - (8n+8) = 8n^2 + 2n - 5 > 0 \qquad (n \geq 1).$$

So by Lemma 3.1 an irreducible module of dimension at most $8n+8$ is trivial, W or the spin module of dimension 2^{2n+1}. For $n \geq 2$ the last is too large: $8n+8 < 2^{2n+1}$ holds at $n = 2$ ($24 < 32$), and if it holds at n, then $8(n+1)+8 = (8n+8)+8 < 2^{2n+1}+8 \leq 4 \cdot 2^{2n+1} = 2^{2(n+1)+1}$. For $n = 1$ the spin module has dimension $8 \leq 16$ and is allowed. □

4. Embeddings of $\mathfrak{so}_m$ in $\mathfrak{so}_{s,t}$

We use three standard facts about a real form $\mathfrak{h}$ of $\mathfrak{so}_m(\mathbb{C})$ with $m = 2l+1$ odd.

(F1) $\mathfrak{h} \cong \mathfrak{so}_{p,q}$ for some $p+q = m$ [7, Ch. X], and $\mathfrak{h}$ is compact exactly when $\{p,q\} = \{m, 0\}$.

(F2) The vector module $\mathbb{R}^{p,q}$ of $\mathfrak{so}_{p,q}$ is absolutely irreducible, since its complexification is W. By Schur's lemma $\mathrm{End}_{\mathfrak{h}}(\mathbb{R}^{p,q}) = \mathbb{R}$, and since $\mathbb{R}^{p,q}$ is isomorphic to its dual through the form of signature (p,q), every $\mathfrak{h}$-invariant bilinear form on $\mathbb{R}^{p,q}$ is a real multiple of that form. The same holds for every absolutely irreducible real module that is isomorphic to its dual.

(F3) If a real $\mathfrak{h}$-module M satisfies $M \otimes \mathbb{C} \cong W^{\oplus c}$, then $M \cong (\mathbb{R}^{p,q})^{\oplus c}$. Indeed the space $\mathrm{Hom}_{\mathfrak{h}}(\mathbb{R}^{p,q}, M) \otimes \mathbb{C} = \mathrm{Hom}_{\mathfrak{h}\otimes\mathbb{C}}(W, M \otimes \mathbb{C})$ is nonzero, so M contains a copy of the irreducible module $\mathbb{R}^{p,q}$; a complement has complexification $W^{\oplus(c-1)}$, and we argue by induction.

Lemma 4.1. *Let $\mathfrak{h} \subset \mathfrak{so}_{s,t}$ be a subalgebra isomorphic to a real form of $\mathfrak{so}_m(\mathbb{C})$, with m odd, and let $V_0 = \{v \in V : \mathfrak{h}v = 0\}$ and $V_1 = \mathfrak{h}V$, the span of all Xv. Then $V = V_1 \oplus V_0$, the two summands are orthogonal, η is nondegenerate on each, and $\mathfrak{h}$ acts faithfully on V_1.*

(a) *If $(s,t) = (8n+3, 3)$, $m = 4n+3$ and $n \geq 2$, then $V_1 \otimes \mathbb{C}$ is isomorphic to the vector module W of $\mathfrak{h} \otimes \mathbb{C}$.*

(b) *If $(s,t) = (8n+4, 4)$, $m = 4n+3$, $n \geq 2$ and $\mathfrak{h}$ is compact, then $V_1 \otimes \mathbb{C} \cong W$ and V_1 is positive definite.*

(c) *If $(s,t) \in \{(11,3), (12,4)\}$ and $\mathfrak{h} \cong \mathfrak{so}_7$ is compact, then V_1 is positive definite of dimension 7 or 8, and $\mathfrak{h}$ acts on it by the vector module or by the spin module.*

Proof. Since $\mathfrak{h}$ is semisimple, V is a direct sum of irreducible $\mathfrak{h}$-modules. For an irreducible summand M, the subspace $\mathfrak{h}M$ is a submodule of M, so it is M if $\mathfrak{h}$ acts nontrivially on M and 0 if M is trivial. Hence V_1 is the sum of the nontrivial irreducible summands, V_0 is the sum of the trivial ones, and $V = V_1 \oplus V_0$. For $x \in V_0$, $X \in \mathfrak{h}$ and $z \in V$, invariance of η gives $\eta(x, Xz) = -\eta(Xx, z) = 0$, so $V_0 \perp V_1$. If η were degenerate on V_1, a nonzero vector of V_1 orthogonal to V_1 would also be orthogonal to V_0, hence to V, which is impossible; the same applies to V_0. Finally $\mathfrak{so}_{s,t}$ acts faithfully on V, $\mathfrak{h}$ acts by zero on V_0, so $\mathfrak{h}$ acts faithfully on V_1, and in particular $V_1 \neq 0$.

Suppose that $V_1 \otimes \mathbb{C} \cong W \oplus W$. By (F1) and (F3), $V_1 \cong \mathbb{R}^{p,q} \otimes \mathbb{R}^2$ with $\mathfrak{h}$ acting on the first factor. Let b be the invariant form on $\mathbb{R}^{p,q}$, of signature (p,q). Every invariant bilinear form on $\mathbb{R}^{p,q} \otimes \mathbb{R}^2$ is $b \otimes M$ for a real 2×2 matrix M, because $\mathrm{Hom}_{\mathfrak{h}}(\mathbb{R}^{p,q} \otimes \mathbb{R}^2, (\mathbb{R}^{p,q} \otimes \mathbb{R}^2)^*) = \mathrm{Hom}_{\mathfrak{h}}(\mathbb{R}^{p,q}, (\mathbb{R}^{p,q})^*) \otimes \mathrm{End}(\mathbb{R}^2)$ and the first factor is $\mathbb{R}b$ by (F2). The restriction of η is symmetric and nondegenerate, so M is symmetric and invertible. After an orthogonal change of basis of

$\mathbb{R}^2$, $M = \operatorname{diag}(\mu_1, \mu_2)$ with $\mu_1\mu_2 \neq 0$, and V_1 is the orthogonal sum of two copies of $(\mathbb{R}^{p,q}, \mu_r b)$. A copy with $\mu_r > 0$ has signature (p, q) and a copy with $\mu_r < 0$ has signature (q, p). Hence

$$\text{the signature of } V_1 \text{ is } (2p, 2q),\ (m, m) \text{ or } (2q, 2p). \tag{9}$$

(a) Here $\dim V = 8n + 6 = 2m$. By Corollary 3.2 the $\mathfrak{h} \otimes \mathbb{C}$-module $V_1 \otimes \mathbb{C}$, which has no trivial summand, is W or $W \oplus W$. In the second case $\dim V_1 = 2m = \dim V$, so $V_1 = V$ has signature $(8n+3, 3)$. This is not $(2p, 2q)$ or $(2q, 2p)$, because $8n+3$ is odd; and it is not (m, m), because $m = 4n + 3 > 3$. So $V_1 \otimes \mathbb{C} \cong W$.

(b) Here $\dim V = 8n+8$, and again $V_1 \otimes \mathbb{C}$ is W or $W \oplus W$. Since $\mathfrak{h}$ is compact, $(p, q) = (m, 0)$ in (F1) and b is definite. In the second case $\dim V_0 = 8n + 8 - 2m = 2$ and, by (9), V_1 has 0, m or $2m$ negative directions; adding the at most 2 negative directions of V_0, the space V would have at most 2 or at least $m = 4n + 3 \geq 11$ negative directions, not 4. So $V_1 \otimes \mathbb{C} \cong W$, and $V_1 \cong \mathbb{R}^m$ with $\eta|_{V_1}$ a nonzero multiple of the definite form b. If it were negative definite, V would have at least $m > 4$ negative directions; so V_1 is positive definite.

(c) By Corollary 3.2 with $n = 1$, $V_1 \otimes \mathbb{C}$ is a sum of copies of $\mathbb{C}^7$ and of the spin module $\mathbb{C}^8$. The compact algebra $\mathfrak{so}_7$ has the real modules $\mathbb{R}^7$ and $\mathbb{R}^8$ (the spin module, realised on the octonions) with these complexifications; both are absolutely irreducible, isomorphic to their duals and carry definite invariant forms, unique up to a factor by (F2). The argument of (F3) shows that V_1 is a sum of copies of $\mathbb{R}^7$ and $\mathbb{R}^8$. Two summands that are not isomorphic are orthogonal, because an invariant pairing between them is an element of $\operatorname{Hom}_{\mathfrak{h}}(M, N^*) = \operatorname{Hom}_{\mathfrak{h}}(M, N) = 0$. The argument that gave (9) shows that c copies of a definite module of dimension d have signature $(c'd, c''d)$ with $c' + c'' = c$. Since $\dim V_1 \leq 16$, the possibilities with at least two summands are $\mathbb{R}^7 \oplus \mathbb{R}^7$, $\mathbb{R}^7 \oplus \mathbb{R}^8$ and $\mathbb{R}^8 \oplus \mathbb{R}^8$, for which the number of negative directions of V_1 lies in $\{0, 7, 14\}$, $\{0, 7, 8, 15\}$ and $\{0, 8, 16\}$ respectively. In each case $\dim V_0 \leq 16 - 14 = 2$, so V would have at most 2 or at least 7 negative directions, never $t \in \{3, 4\}$. Hence V_1 is a single copy of $\mathbb{R}^7$ or $\mathbb{R}^8$, with a definite form; a negative definite form would give at least $7 > t$ negative directions, so V_1 is positive definite. □

5. Proofs of Theorems A and B

Lemma 5.1. *In the situation of Lemma 4.1(a) or (b), put $U = V_1 \otimes \mathbb{C}$. Then U is a nondegenerate subspace of $V_{\mathbb{C}}$ of dimension m, $U^\perp = V_0 \otimes \mathbb{C}$, and the image of $\mathfrak{h} \otimes \mathbb{C}$ in $\mathfrak{so}(V_{\mathbb{C}})$ is exactly $\mathfrak{so}(U)$, the algebra of maps that preserve $\eta|_U$ and vanish on $U^\perp$. Under this identification the complexified vector module of $\mathfrak{h}$ is U, and Λ^k of it is $\Lambda^k U$.*

Proof. By Lemma 4.1, $V = V_1 \oplus V_0$ orthogonally with η nondegenerate on both, so the same holds for $V_{\mathbb{C}} = U \oplus (V_0 \otimes \mathbb{C})$, and $U^\perp = V_0 \otimes \mathbb{C}$. Every $X \in \mathfrak{h} \otimes \mathbb{C}$ maps U into U, vanishes on $V_0 \otimes \mathbb{C}$, and is skew for η; so $X \in \mathfrak{so}(U)$. The map $\mathfrak{h} \otimes \mathbb{C} \to \mathfrak{so}(U)$ is injective because $\mathfrak{h}$ acts faithfully on V_1. Both sides have dimension $m(m-1)/2$, so it is an isomorphism. Finally $U = V_1 \otimes \mathbb{C}$ is, by Lemma 4.1, the complexified vector module of $\mathfrak{h}$. □

Proof of Theorem A. By Section 2 it suffices to show that $\operatorname{Hom}_{\mathfrak{h}\otimes\mathbb{C}}(\Lambda^k U, S^\pm) = 0$ for $0 \leq k \leq m$, with U as in Lemma 5.1. By that lemma, $\mathfrak{h} \otimes \mathbb{C}$ acts on $\Lambda^k U$ and on $S^\pm$ through $\mathfrak{so}(U) \subset \mathfrak{so}(V_{\mathbb{C}})$, so the Casimir element Ω of (6), built from an orthonormal basis $u_1, \ldots, u_m$ of U, acts on both. By Lemma 2.1(i) it acts on $\Lambda^k U$ by the integer $k(m-k)$, and by Lemma 2.1(ii) it acts on S^+ and S^- by $m(m-1)/8$; the action on S is given by (4) applied to the vectors $u_i \in U \subset V_{\mathbb{C}}$, so it is the restriction of the spin representation of $\mathfrak{so}(V_{\mathbb{C}})$. If $f : \Lambda^k U \to S^\pm$ is a module map, then by Lemma 2.1(o)

$$\tfrac{m(m-1)}{8}\, f(x) = \Omega f(x) = f(\Omega x) = k(m-k)\, f(x) \qquad (x \in \Lambda^k U),$$

and since $m(m-1)/8 \neq k(m-k)$ by Lemma 2.1(iii) with $m = 4n + 3$, we get $f = 0$. □

For $n = 3$ this compares $105/4 = 26.25$ on the spinors of $\mathfrak{so}_{27,3}$ with the values 0, 14, 26, 36, 44, 50, 54 and 56 on the exterior powers of **15**; the closest value, 26 for $\Lambda^2\mathbf{15} = \mathbf{105}$ and $\Lambda^{13}\mathbf{15}$, still differs. In the language of [12]: under any $\mathfrak{so}_{15}$ inside $\mathfrak{so}_{27,3}$ the module **16384** is 128 copies of the spin module **128** of $\mathfrak{so}_{15}$, and it contains none of the modules **15**, **455**, **3003**, **6435**, **5005**, **1365**, **105**, **1** of (2). Indeed, by Lemma 2.1(o),(ii) every irreducible summand of $S^{\pm}$ has Casimir value $105/4$; choose a vector $u_0 \in U^{\perp}$ with $\eta(u_0, u_0) = 1$ and put $U' = U \oplus \mathbb{C}u_0$, of dimension 16. Then $S \cong S_{U'} \otimes S_{U'^{\perp}}$ as a module over $\mathfrak{so}(U') \oplus \mathfrak{so}(U'^{\perp})$ [9, Ch. I, §1], and each semispinor of $\mathfrak{so}(U') \cong \mathfrak{so}_{16}(\mathbb{C})$ restricts to the spin module of $\mathfrak{so}(U) \cong \mathfrak{so}_{15}(\mathbb{C})$ [6, §20.1], so $\mathfrak{so}(U)$ acts on S through copies of its spin module; the count $16384/128 = 128$ follows.

Proof of Theorem B. By Lemma 4.1(c) there are two cases.

Case 1: $V_1 \cong \mathbb{R}^7$, with $\mathfrak{h}$ acting by the vector module. The proof of Lemma 5.1 applies verbatim with $m = 7$, and the proof of Theorem A then shows that Ω acts by $21/4$ on $S^{\pm}$ and by $k(7-k) \in \{0, 6, 10, 12\}$ on $\Lambda^k\mathbb{C}^7$, so no exterior power of $\mathbb{C}^7$ occurs in $S^{\pm}$.

Case 2: $V_1 \cong \mathbb{R}^8$, with $\mathfrak{h}$ acting by the spin module. Put $U = V_1 \otimes \mathbb{C}$, a nondegenerate subspace of dimension 8, and $U^{\perp} = V_0 \otimes \mathbb{C}$, of dimension $2j$ with $j = 3$ for $\mathfrak{so}_{11,3}$ and $j = 4$ for $\mathfrak{so}_{12,4}$. As in Lemma 5.1, the image $\mathfrak{h}_{\mathbb{C}}$ of $\mathfrak{h} \otimes \mathbb{C}$ lies in $\mathfrak{so}(U) \cong \mathfrak{so}_8(\mathbb{C})$ and is zero on $U^{\perp}$; it is the image of $\mathfrak{so}_7(\mathbb{C})$ under its spin representation on U.

Step 1: the standard $\mathfrak{so}_7$. Fix $e \in U$ with $\eta(e, e) = 1$ and let $\mathfrak{k} = \mathfrak{so}(e^{\perp} \cap U) \cong \mathfrak{so}_7(\mathbb{C})$, the stabiliser of e in $\mathfrak{so}(U)$. On $U = (e^{\perp} \cap U) \oplus \mathbb{C}e$ it acts by $\mathbb{C}^7 \oplus \mathbb{C}$. On each semispinor $S_U^{\pm}$ of $\mathfrak{so}(U)$, which has dimension 8, Lemma 2.1(ii) applied to the 7-dimensional subspace $e^{\perp} \cap U$ shows that the Casimir element of $\mathfrak{k}$ acts by $7 \cdot 6/8 = 21/4$. By Corollary 3.2 with $n = 1$, an 8-dimensional $\mathfrak{so}_7(\mathbb{C})$-module is a sum of trivial modules (Casimir 0), copies of $\mathbb{C}^7$ (Casimir 6) and copies of $\mathbb{C}^8$ (Casimir $21/4$). Hence $\mathfrak{k}$ acts on S_U^+ and on S_U^- by the spin module $\mathbb{C}^8$.

Step 2: triality. The three 8-dimensional irreducible modules U, S_U^+, S_U^- of $\mathfrak{so}(U)$ are permuted by the automorphisms of $\mathfrak{so}(U)$, and every permutation of the three occurs [6, §20.3]. Choose an automorphism τ with $U \circ \tau \cong S_U^+$, where $R \circ \tau$ denotes the module R with X acting by $\tau(X)$. Then $\tau(\mathfrak{k})$ acts on U as $\mathfrak{k}$ acts on S_U^+, that is, by the spin module, through an embedding $\mathfrak{so}_7(\mathbb{C}) \to \mathfrak{so}(U)$. The algebra $\mathfrak{h}_{\mathbb{C}}$ is also the image of an embedding $\mathfrak{so}_7(\mathbb{C}) \to \mathfrak{so}(U)$ by the spin module. Two such embeddings differ by an isomorphism of U intertwining the two actions; since the spin module of $\mathfrak{so}_7(\mathbb{C})$ carries a unique invariant symmetric form up to a factor, a suitable multiple g of that isomorphism lies in $\mathrm{O}(U)$, and $\mathfrak{h}_{\mathbb{C}} = g\,\tau(\mathfrak{k})\,g^{-1}$. Put $\varphi = \mathrm{Ad}(g) \circ \tau$, an automorphism of $\mathfrak{so}(U)$ with $\varphi(\mathfrak{k}) = \mathfrak{h}_{\mathbb{C}}$. Since $\mathrm{Ad}(g)$ fixes the class of U and permutes $\{S_U^+, S_U^-\}$, the automorphism φ permutes the three classes with $U \circ \varphi \cong S_U^+$; hence $\{S_U^+ \circ \varphi,\ S_U^- \circ \varphi\} = \{U,\ S_U^-\}$. For any module R of $\mathfrak{so}(U)$, the action of $\mathfrak{h}_{\mathbb{C}} = \varphi(\mathfrak{k})$ on R is the action of $\mathfrak{k}$ on $R \circ \varphi$. By Step 1, $\mathfrak{k}$ acts on U by $\mathbb{C}^7 \oplus \mathbb{C}$ and on S_U^- by $\mathbb{C}^8$. Therefore $\mathfrak{h}_{\mathbb{C}}$ acts on U by $\mathbb{C}^8$, and on S_U^+, S_U^- by $\mathbb{C}^7 \oplus \mathbb{C}$ and $\mathbb{C}^8$, in some order.

Step 3: the semispinors of $\mathfrak{so}_{s,t}$. Since $V_{\mathbb{C}} = U \oplus U^{\perp}$ orthogonally with both summands of even dimension, $\mathrm{Cl}(V_{\mathbb{C}})$ is the graded tensor product of $\mathrm{Cl}(U)$ and $\mathrm{Cl}(U^{\perp})$, $S \cong S_U \otimes S_{U^{\perp}}$, and the chirality operator of $V_{\mathbb{C}}$ is the product of those of U and $U^{\perp}$ [9, Ch. I, §1]. Hence

$$S^+ \cong S_U^+ \otimes S_{U^{\perp}}^+ \oplus S_U^- \otimes S_{U^{\perp}}^-, \qquad S^- \cong S_U^+ \otimes S_{U^{\perp}}^- \oplus S_U^- \otimes S_{U^{\perp}}^+$$

as modules over $\mathfrak{so}(U) \oplus \mathfrak{so}(U^{\perp})$, with $\dim S_{U^{\perp}}^{\pm} = 2^{j-1}$. The algebra $\mathfrak{h}_{\mathbb{C}}$ lies in $\mathfrak{so}(U)$, so it acts on the first factors only. By Step 2, each of S^+ and S^- is, as an $\mathfrak{h}_{\mathbb{C}}$-module, 2^{j-1} copies of $\mathbb{C} \oplus \mathbb{C}^7 \oplus \mathbb{C}^8$; the dimensions agree, $2^{j-1} \cdot 16 = 2^{j+3} = \dim S^{\pm}$.

Step 4: Casimir values. Every automorphism of $\mathfrak{so}_7(\mathbb{C})$ is inner and Ω is central, so the Casimir element of an algebra isomorphic to $\mathfrak{so}_7(\mathbb{C})$ does not depend on the isomorphism. The Casimir element of $\mathfrak{h}_{\mathbb{C}}$ acts on the summands $\mathbb{C}$, $\mathbb{C}^7$, $\mathbb{C}^8$ by 0, 6 and $21/4$ (Lemma 2.1 with

$m = 7$, as in Step 1), and on $\Lambda^k\mathbb{C}^7$ by $k(7-k)$, which is 10 for $k = 2, 5$ and 12 for $k = 3, 4$. By Lemma 2.1(o) there is no nonzero module map from $\Lambda^k\mathbb{C}^7$ to $S^{\pm}$ for $k = 2, 3, 4, 5$.

In both cases none of $\Lambda^2\mathbb{R}^7, \Lambda^3\mathbb{R}^7, \Lambda^4\mathbb{R}^7, \Lambda^5\mathbb{R}^7$ occurs in $S^{\pm}$, even after complexification. Each of $\bigoplus_k \Lambda^k\mathbb{R}^7$, $\bigoplus_i \Lambda^{2i}\mathbb{R}^7$ and $\bigoplus_i \Lambda^{2i+1}\mathbb{R}^7$ contains one of them, which proves the last statement. □

In Case 2 the modules Λ^0, Λ^1, Λ^6 and Λ^7 of $\mathbb{C}^7$ do occur in $S^{\pm}$; this is why Theorem B lists only the four middle powers.

Remark 5.2. Compactness in Theorem A(b) cannot be dropped. Let $p = 4n + 1$, so that $m = p + 2$, and let $\mathfrak{h} = \mathfrak{so}_{p,2}$ act diagonally on $\mathbb{R}^{p,2} \oplus \mathbb{R}^{p,2}$ and trivially on $\mathbb{R}^{2,0}$, inside $V = \mathbb{R}^{p,2} \oplus \mathbb{R}^{p,2} \oplus \mathbb{R}^{2,0}$, of signature $(2p + 2, 4) = (8n + 4, 4)$. Over $\mathbb{C}$, the space $U = \mathbb{C}^m \oplus \mathbb{C}^m$ (with the form $b \oplus b$) contains the subspaces $L_{\pm} = \{(u, \pm iu) : u \in \mathbb{C}^m\}$. They are isotropic, since $b(u, u') + (\pm i)^2 b(u, u') = 0$; they are $\mathfrak{h}$-stable and isomorphic to $\mathbb{C}^m$; and $U = L_+ \oplus L_-$. So $\mathfrak{h} \otimes \mathbb{C} \subset \mathfrak{gl}(L_+)$, and the semispinors of $\mathfrak{so}(U)$ are $\Lambda^{\mathrm{even}}L_+$ and $\Lambda^{\mathrm{odd}}L_+$ twisted by the square root of the inverse determinant of $\mathfrak{gl}(L_+)$, which is trivial on $\mathfrak{so}_m$ [6, §20.1]. Tensoring with the two one-dimensional semispinors of $\mathbb{R}^{2,0} \otimes \mathbb{C}$, as in Step 3 above, shows that each semispinor of $\mathfrak{so}_{8n+4,4}$ restricts to $\Lambda^{\mathrm{even}}\mathbb{C}^m \oplus \Lambda^{\mathrm{odd}}\mathbb{C}^m = \Lambda^{\bullet}\mathbb{C}^m$. The algebra $\mathfrak{so}_{p,2}$ is not the rotation algebra of a spatial worldvolume. In $\mathfrak{so}_{8n+3,3}$ this construction is excluded by Lemma 4.1(a): there $V_1 = V$ would have signature $(8n + 3, 3)$ with an odd entry.

Remark 5.3. The argument says nothing against the algebras $\mathfrak{e}^{(n)}_{8(-24)} = \mathfrak{so}_{8n+4,4} \oplus \mathbf{2}^{4n+3}$ of [14] or their grading; it shows only that their spinor generators are not exterior algebras over the rotations of a brane. Other parts of Section III of [12] contain smaller slips: the sums in (3.4) and (3.5) run up to $i = 15$ instead of $i = 7$, and the last term of (3.8) contracts a gamma matrix with $n + 3$ indices against a $(4n + 3)$-form.

6. The dual M-theory

We use the conventions of [12]. In a spacetime of signature $s + t$, a p-brane carries a p-form charge and has a worldvolume of signature $p + t$: the 11-brane of $27 + 3$ has worldvolume $11 + 3$, the 10-brane of $26 + 2$ has worldvolume $10 + 2$, and the 10-brane of $26 + 1$ has worldvolume $10 + 1$. In a spacetime of dimension D with one time, a p-brane couples to a $(p + 1)$-form potential with a $(p + 2)$-form field strength, whose Hodge dual has degree $D - p - 2$; so the magnetic dual of a p-brane is a $(D - p - 4)$-brane, as for the 2-brane and the 21-brane of bosonic M-theory in $26 + 1$ dimensions [8]. The pairs $11 \leftrightarrow 15$ in $27 + 3$ and $10 \leftrightarrow 14$ in $26 + 2$ quoted in [12] also satisfy $p + p' = D - 4$.

Proof of Proposition C. The reduction $26 + 2 \to 26 + 1$ removes one timelike direction and no spacelike one. In the conventions above it takes the 10-brane with worldvolume $10 + 2$ to the 10-brane with worldvolume $10 + 1$, as [12] states, and by the same rule it takes the 14-brane with worldvolume $14 + 2$ to a 14-brane with worldvolume $14 + 1$. Passing from $14 + 2$ to $13 + 1$ would remove a spacelike direction of the worldvolume, which the reduction does not do. Moreover, in $D = 27$ the magnetic dual of a 10-brane is a $27 - 10 - 4 = 13$-brane, while $10 + 14 = 24 \neq 23$, so the reduced 14-brane is not the dual of the reduced 10-brane. The "dual M-theory" on a $13 + 1$ worldvolume is therefore not the reduction of the 14-brane of $26 + 2$. □

The reduction $27 + 3 \to 26 + 2$ removes one spacelike and one timelike direction, and there the pairing is preserved: the 11-brane and the 15-brane wrapping the removed spacelike direction become the 10-brane and the 14-brane, and $10 + 14 = 28 - 4$. The pairing fails only at the last step.

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